

ODYSSEY

Smart Hospital Patient Prioritization System Using **Binary Heap**

Louis Christian Dharma Panna Wibowo
2902731990

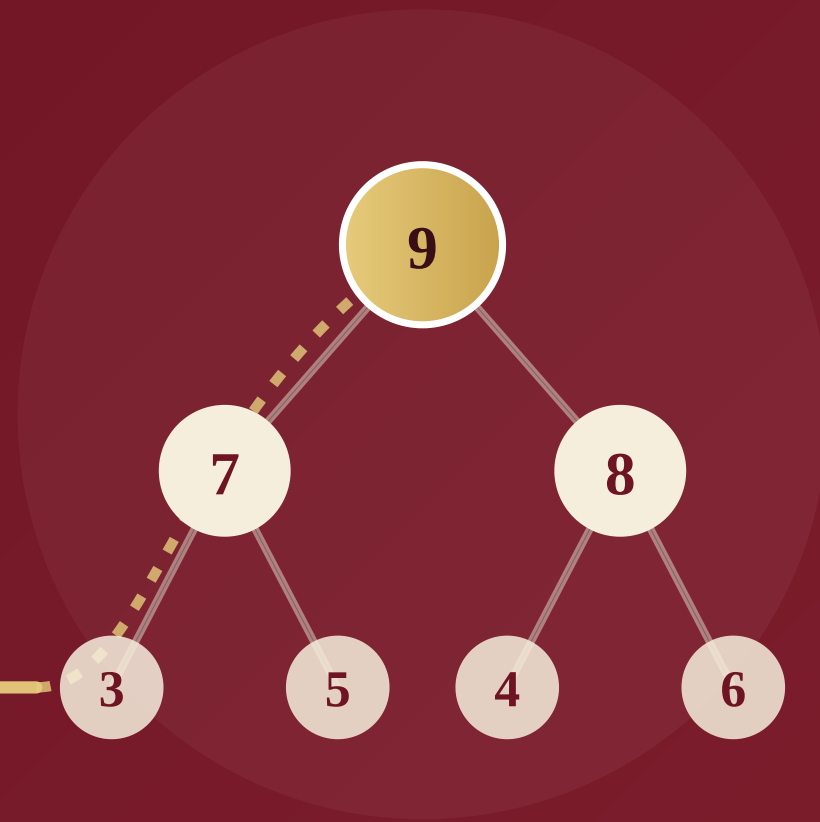
Nizar Ramadhan
2902725672

Kayla Isabelle Limaran
2902728081



PATIENT VITALS

MAX-HEAP · SEVERITY FIRST



1 PROBLEM & OBJECTIVE



Emergency rooms run on minutes. Manual triage struggles to scale under pressure.

PROBLEM

Hospitals prioritize patients **MANUALLY** — causing delays, inefficiency, and human error in emergencies.



SOLUTION

A digital queue that **AUTO-PRIORITIZES** patients by symptom severity in real time.

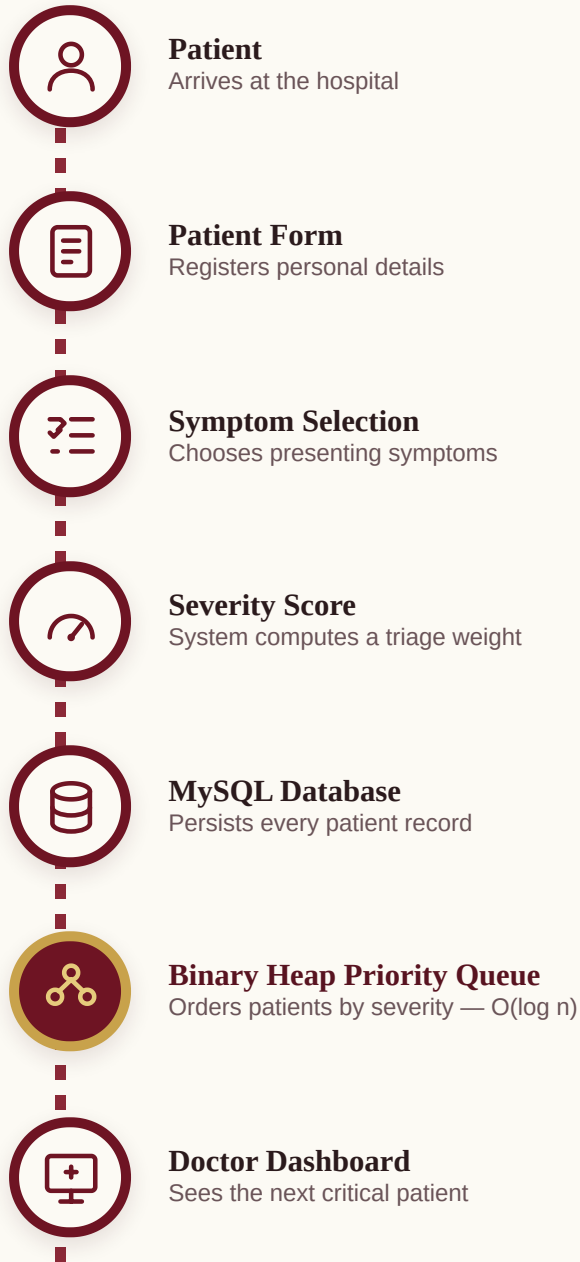
Objective

- Develop a hospital queue-management system that automatically ranks patients according to **symptom severity score**, removing manual bottlenecks.

Severity Score → Priority



2 SYSTEM WORKFLOW



3 DATA STRUCTURE ANALYSIS

Linked List

- $O(1)$ front insertion
- $O(n)$ middle / end insertion

FAST FRONT

Binary Heap

- $O(\log n)$ insertion
- $O(\log n)$ removal (extract-max)
- ★ best overall scalability

SELECTED SOLUTION

ArrayList

- $O(n)$ insertion
- $O(n)$ requires element shifting

SHIFTING

4 PERFORMANCE RESULTS

Insertion benchmark · 100,000 patient records

Front Insertion

Inserting at the head of the queue

LINEAR · MS



Middle Insertion

Inserting into the middle of the queue

LOG SCALE · MS



End Insertion

Inserting at the tail of the queue

LOG SCALE · MS

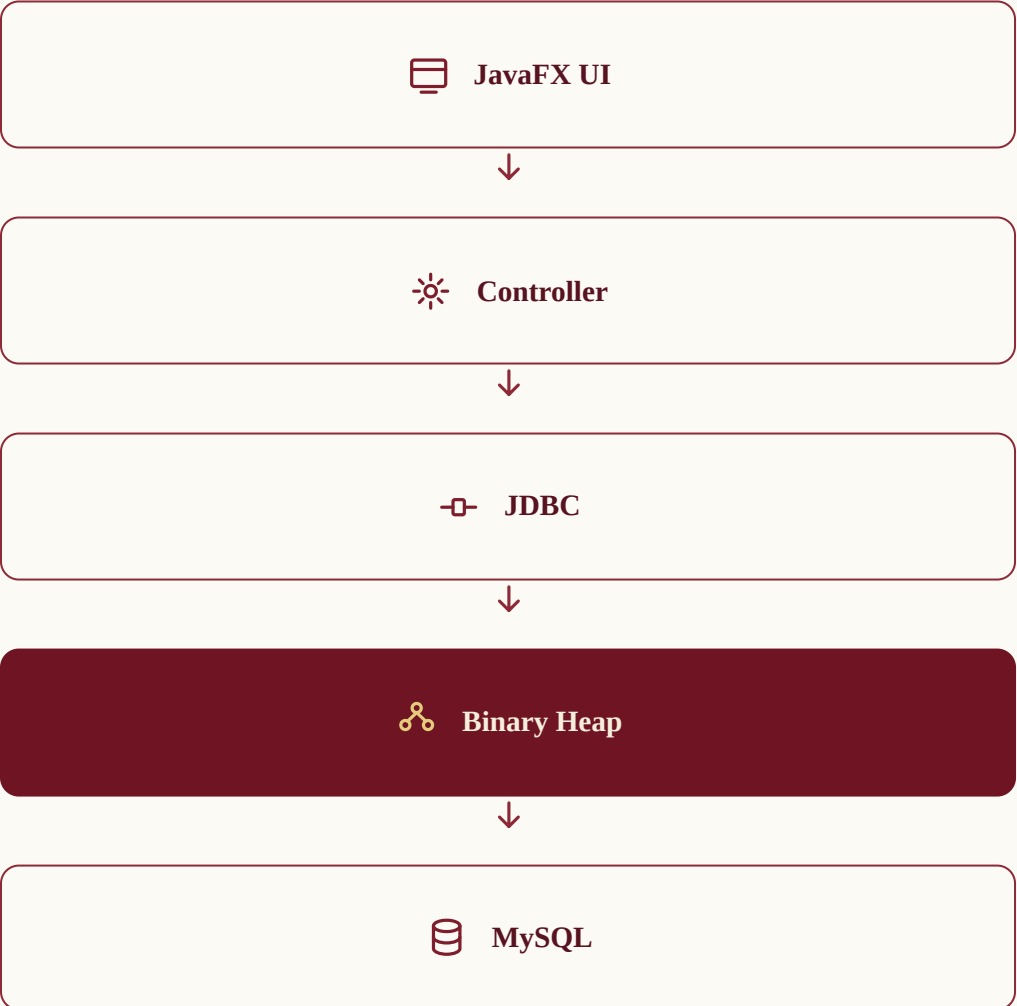


Binary Heap Linked List ArrayList ≈ 600× faster for middle & end insertion



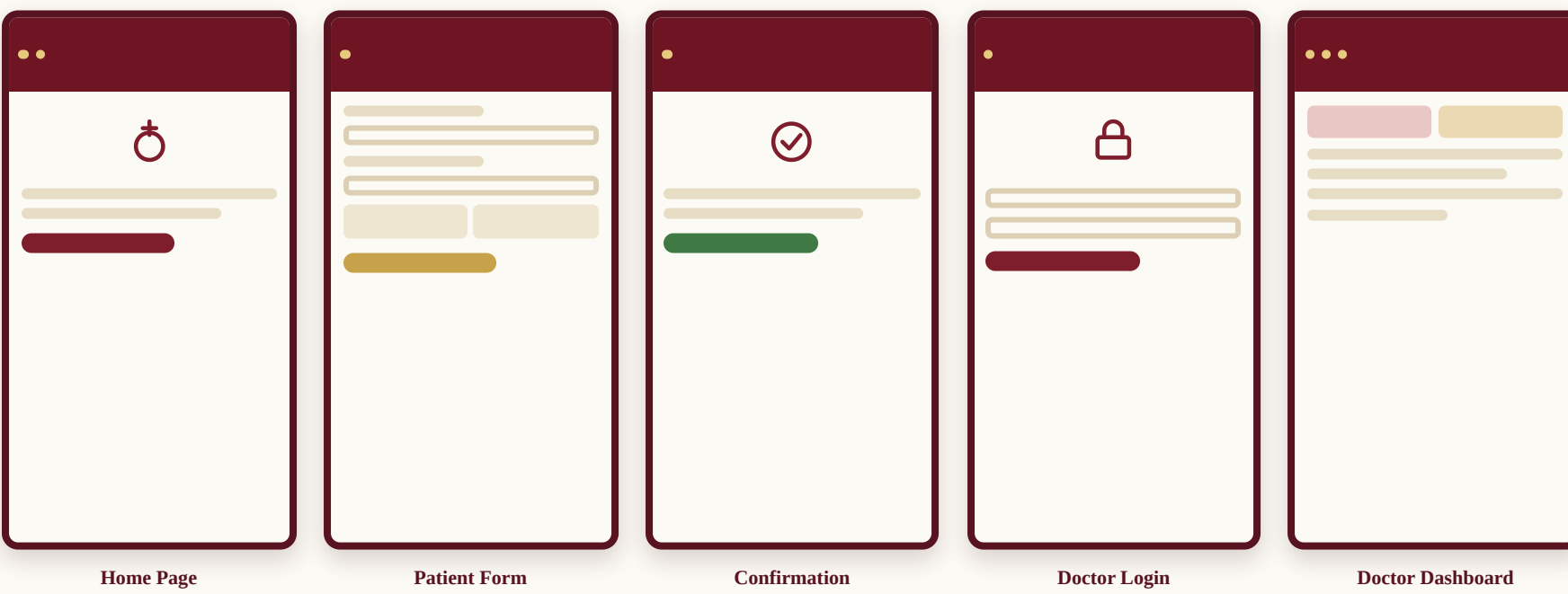
Binary Heap demonstrated the most stable and scalable performance for large patient datasets.

5 SYSTEM ARCHITECTURE

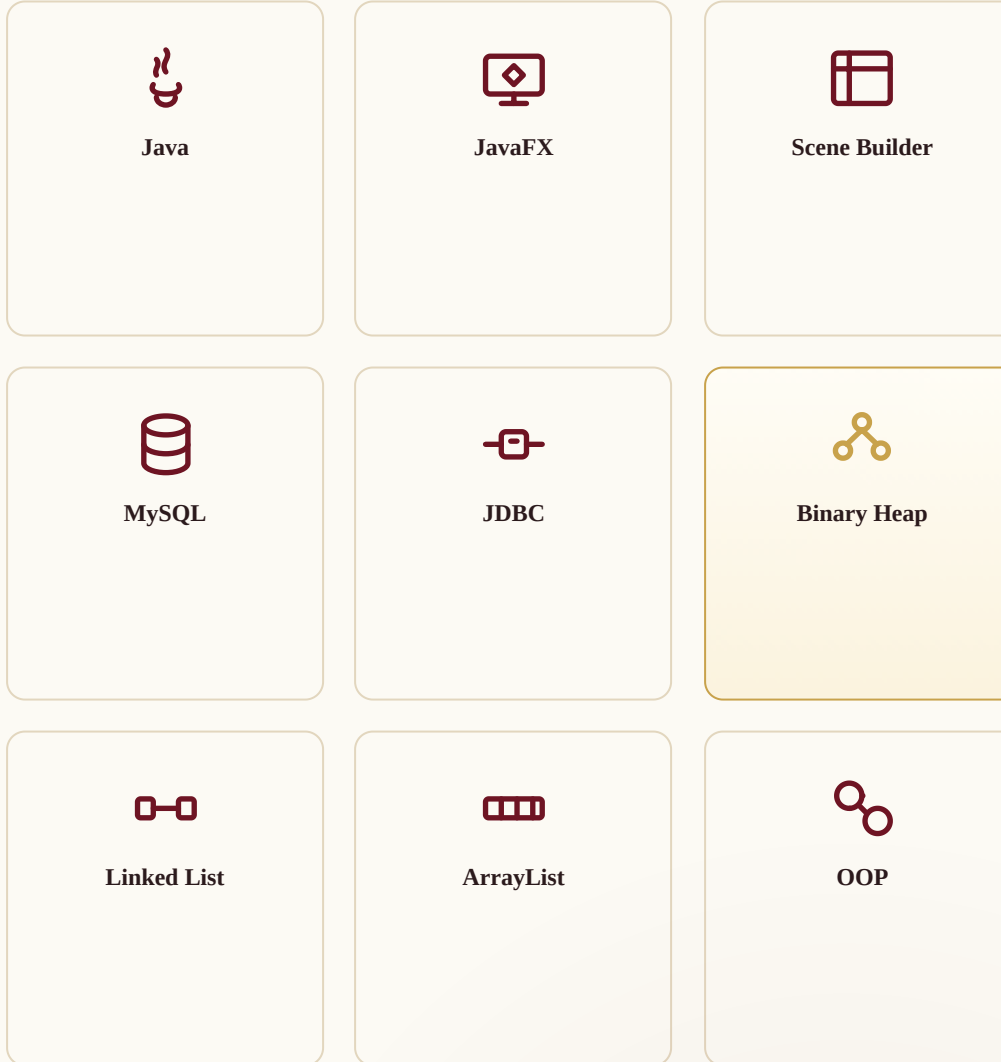


UML layered architecture — separation of UI, logic & data

6 APPLICATION SHOWCASE



7 TECHNOLOGY STACK



CONCLUSION

The **Odyssey** system successfully applies Data Structures and Object-Oriented Programming principles to improve hospital patient prioritization. Experimental results show that the **Binary Heap** consistently provides the best scalability and efficiency, making it the most suitable structure for emergency patient management systems.



Binary Heap is the optimal data structure for real-time hospital patient prioritization.